

FORMAN CHRISTIAN COLLEGE

(A CHARTERED UNIVERSITY)



CSCS 495 A

FA24

TryHackMe

Assignment 3 Report

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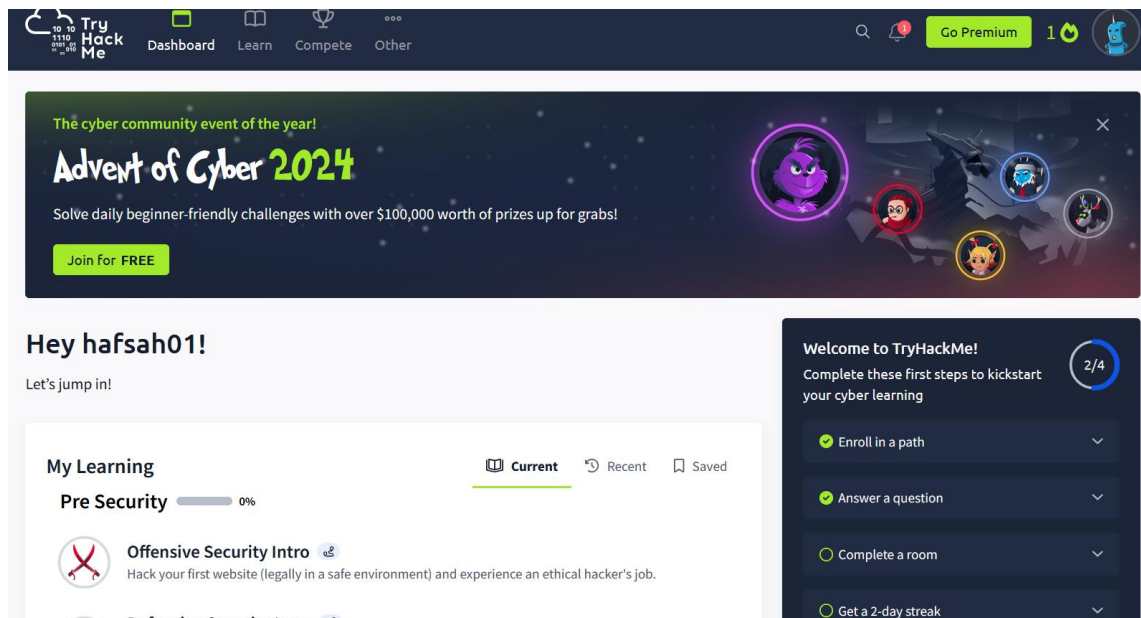
24th November 2024

Introduction

This task involves setting up a TryHackMe account and configuring a VPN connection using OpenVPN. The VPN allows secure connectivity to TryHackMe's virtual network, enabling hands-on cybersecurity exercises. This report details the steps taken, including account creation, VPN configuration, and connectivity verification.

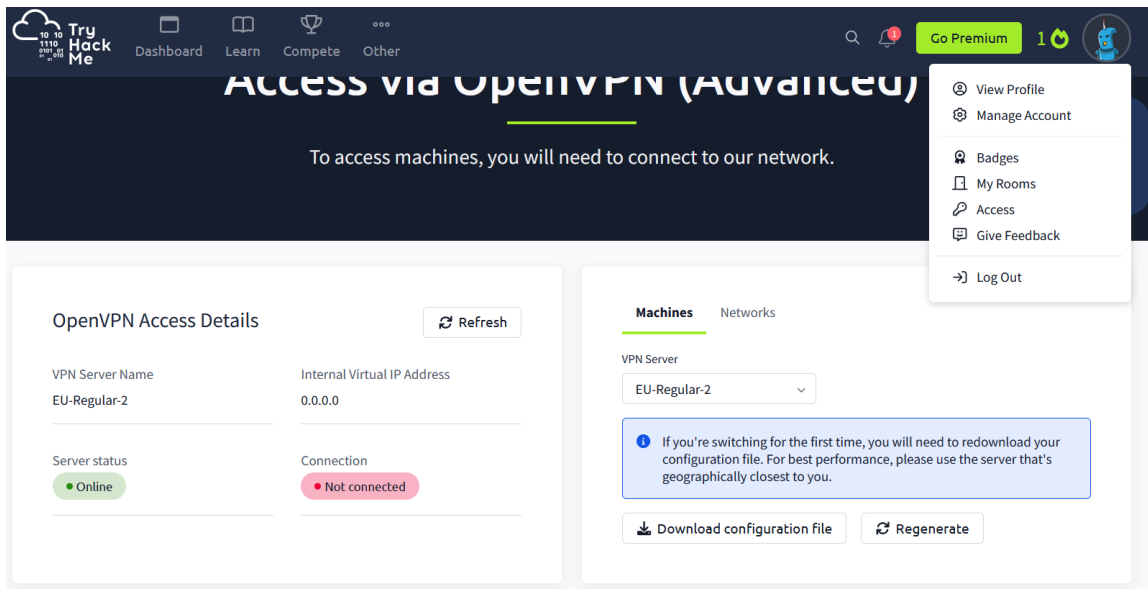
1. Creating a TryHackMe Account

1. Navigate to [TryHackMe](https://tryhackme.com) by searching on Browser in Kali.
2. Click on the "Join For Free" button on the top-right corner.
3. Fill out the required information form with my email address.
4. Verified the email through the confirmation link received.
5. Logged into the TryHackMe portal, landing on the dashboard page.

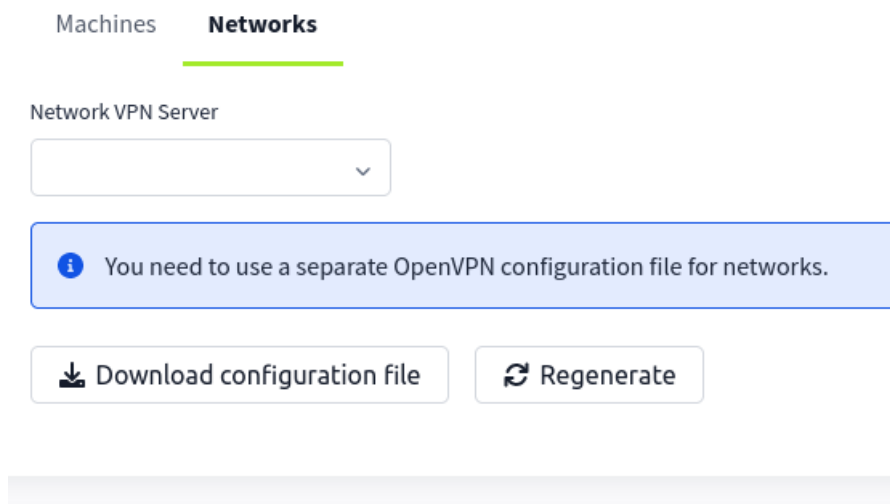


2. Downloading the Configuration File

1. Click on the profile icon at the top-right corner of the dashboard and selected **Access**.

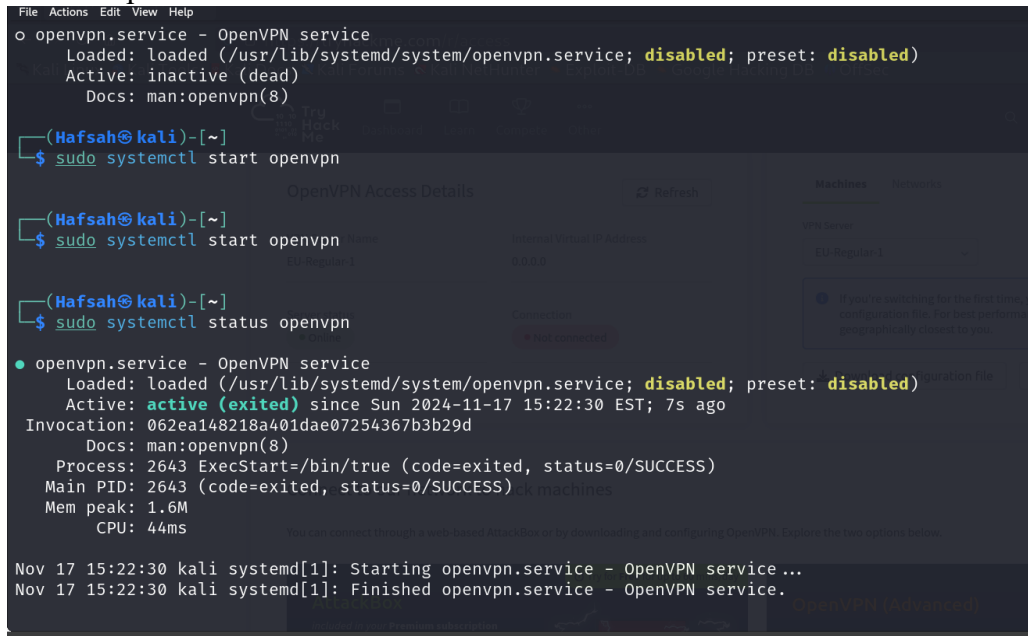


2. Located the option to download the OpenVPN configuration file (.ovpn) and downloaded it.
 - The .ovpn file contains all necessary parameters for establishing the VPN connection.



3. Configuring OpenVPN

1. Opened a terminal in Kali Linux.
2. Ensure OpenVPN was enabled



The screenshot shows a Kali Linux terminal window with the following commands and output:

```
(Hafsah@kali)-[~]
$ sudo systemctl start openvpn
(Hafsah@kali)-[~]
$ sudo systemctl start openvpn
(Hafsah@kali)-[~]
$ sudo systemctl status openvpn
```

The output of the status command shows that the `openvpn.service` is loaded but inactive (dead). It also shows the process details for the service.

Below the terminal window, there is a screenshot of the OpenVPN GUI. It displays the 'OpenVPN Access Details' for a server named 'EU-Reggie-1'. The status is 'Not connected'. A message indicates that if you're switching for the first time, you should check the configuration file for best performance.

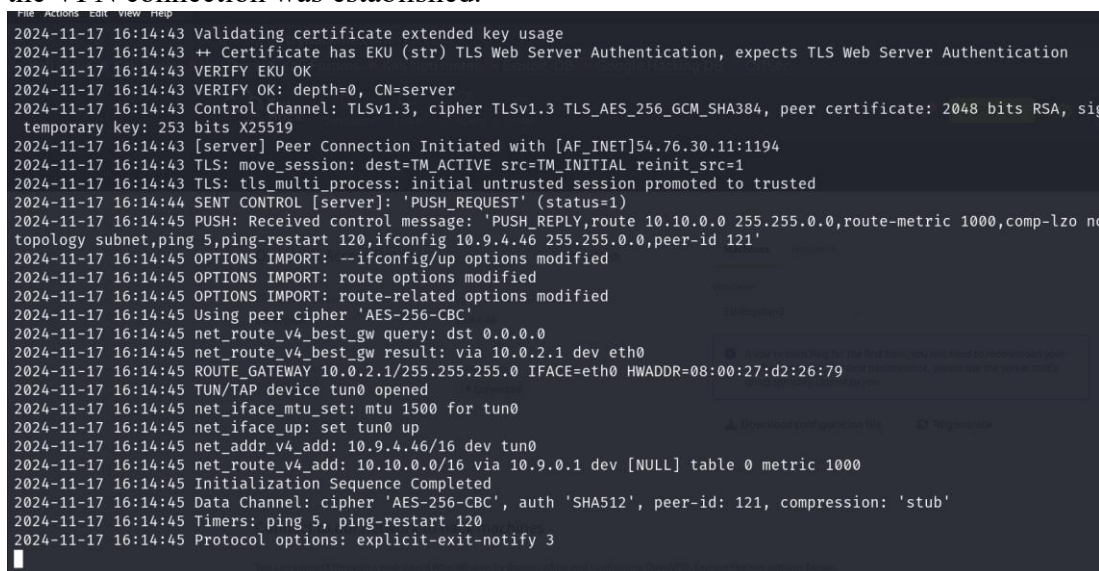
3. Navigated to the directory where the .ovpn file was saved:

```
cd ~/Downloads
```

4. Started OpenVPN with the following command:

```
sudo openvpn hafsah01ovpn
```

The process initialized, and the message Initialization Sequence Completed confirmed the VPN connection was established.



The screenshot shows a terminal window with the following log output:

```
2024-11-17 16:14:43 Validating certificate extended key usage
2024-11-17 16:14:43 ++ Certificate has EKU (str) TLS Web Server Authentication, expects TLS Web Server Authentication
2024-11-17 16:14:43 VERIFY OK
2024-11-17 16:14:43 VERIFY OK: depth=0, CN=server
2024-11-17 16:14:43 Control Channel: TLSv1.3, cipher TLSv1.3 TLS_AES_256_GCM_SHA384, peer certificate: 2048 bits RSA, sig
temporary key: 253 bits X25519
2024-11-17 16:14:43 [server] Peer Connection Initiated with [AF_INET]54.76.30.11:1194
2024-11-17 16:14:43 TLS: move_session: dest=TM_ACTIVE src=TM_INITIAL reinit_src=1
2024-11-17 16:14:43 TLS: tls_multi_process: initial untrusted session promoted to trusted
2024-11-17 16:14:44 SENT CONTROL [server]: 'PUSH_REQUEST' (status=1)
2024-11-17 16:14:45 PUSH: Received control message: 'PUSH_REPLY,route 10.10.0.0 255.255.0.0,route-metric 1000,comp-lzo no
topology subnet,ping 5,ping-restart 120,ifconfig 10.9.4.46 255.255.0.0,peer-id 121'
2024-11-17 16:14:45 OPTIONS IMPORT: --ifconfig/up options modified
2024-11-17 16:14:45 OPTIONS IMPORT: route options modified
2024-11-17 16:14:45 OPTIONS IMPORT: route-related options modified
2024-11-17 16:14:45 Using peer cipher 'AES-256-CBC'
2024-11-17 16:14:45 net_route_v4_best_gw query: dst 0.0.0.0
2024-11-17 16:14:45 net_route_v4_best_gw result: via 10.0.2.1 dev eth0
2024-11-17 16:14:45 ROUTE_GATEWAY 10.0.2.1/255.255.255.0 IFACE=eth0 HWADDR=08:00:27:d2:26:79
2024-11-17 16:14:45 TUN/TAP device tun0 opened
2024-11-17 16:14:45 net_iface_mtu_set: mtu 1500 for tun0
2024-11-17 16:14:45 net_iface_up: set tun0 up
2024-11-17 16:14:45 net_addr_v4_add: 10.9.4.46/16 dev tun0
2024-11-17 16:14:45 net_route_v4_add: 10.10.0.0/16 via 10.9.0.1 dev [NULL] table 0 metric 1000
2024-11-17 16:14:45 Initialization Sequence Completed
2024-11-17 16:14:45 Data Channel: cipher 'AES-256-CBC', auth 'SHA512', peer-id: 121, compression: 'stub'
2024-11-17 16:14:45 Timers: ping 5, ping-restart 120
2024-11-17 16:14:45 Protocol options: explicit-exit-notify 3
```

4. Verifying the VPN Connection

1. Opened a new terminal tab and checked network interfaces:

Ifconfig

Confirmed the creation of the tun0 interface with an assigned IP address, indicating the VPN was active.

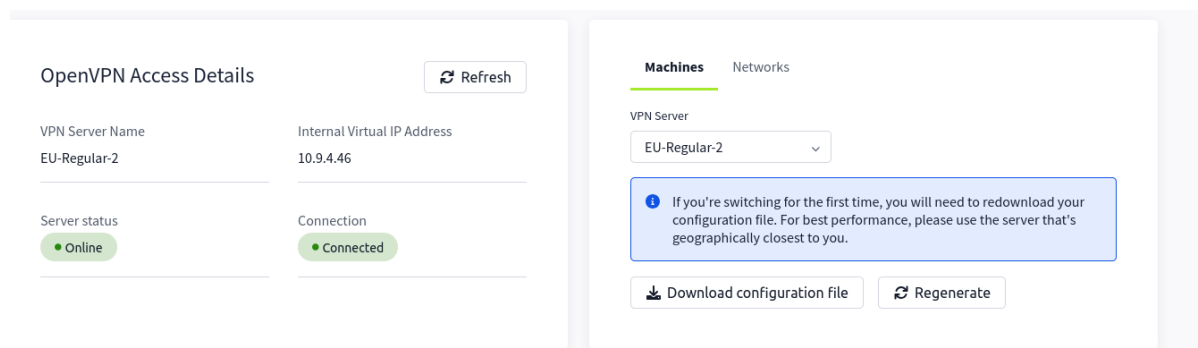
```
(Hafsa@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.4 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::ed79:3165:21d:8bcf prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:d2:26:79 txqueuelen 1000 (Ethernet)
    RX packets 1472 bytes 562122 (548.9 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1812 bytes 307185 (299.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

tun0: flags=4305<UP,POINTOPOINT,RUNNING,NOARP,MULTICAST> mtu 1500
    inet 10.8.71.185 netmask 255.255.0.0 destination 10.8.71.185
    inet6 fe80::20b5:2a60:d628:44e3 prefixlen 64 scopeid 0x20<link>
    unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00 txqueuelen 500 (UNSPEC)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 4 bytes 192 (192.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

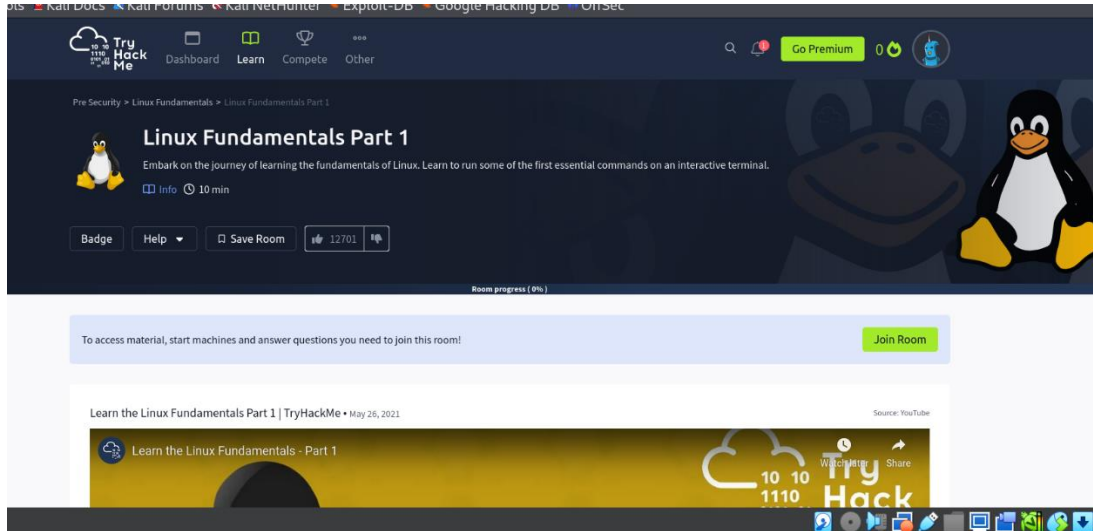
2. Returned to the TryHackMe Access page and refreshed. Verified:

- VPN Server Name
- Internal Virtual IP Address
- Server Status: Online
- Connection: Connected

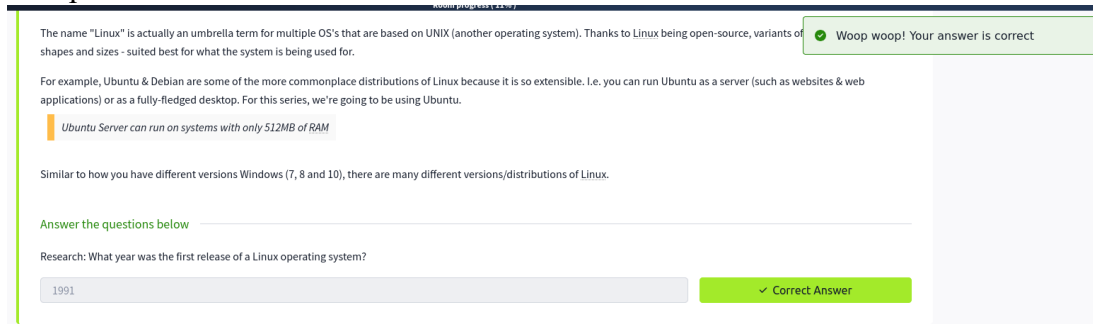


5. Exploring TryHackMe Linux Room

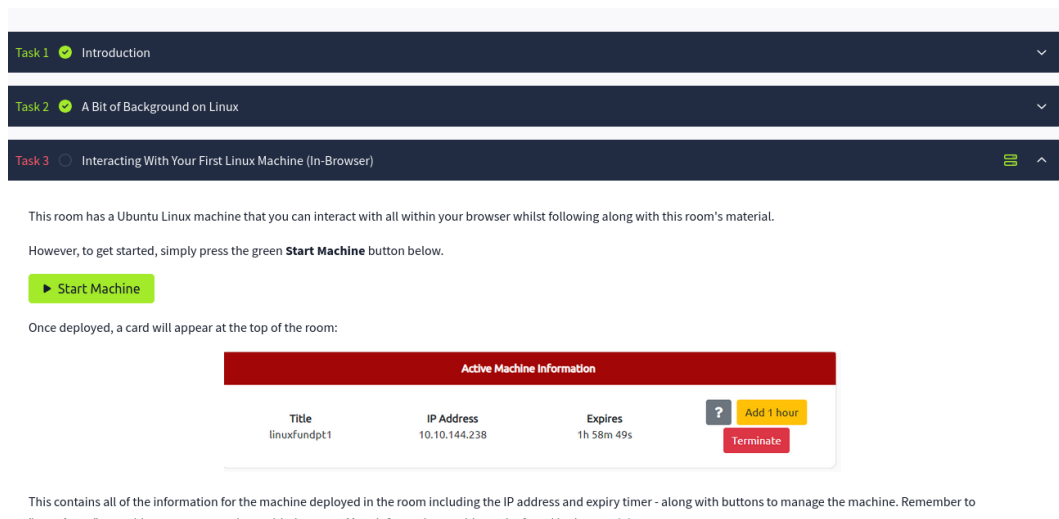
1. Navigated to the **Learn** section and searched for "Linux".
2. Selected the first room titled **Linux Fundamentals Part 1**.

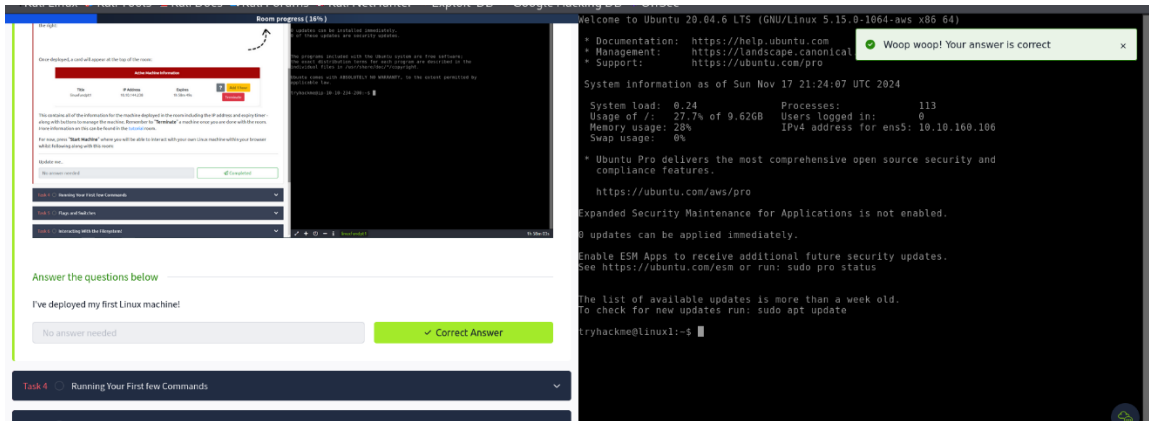


3. Completed Task 1 and 2



4. Started the machine for Task 3 in the room and verified connectivity with the target IP address.





5. Practiced commands and concepts on the provided Linux terminal.

Title	Target IP Address	Expires	
linuxfundpart1v2	10.10.160.106	54min 35s	Add 1 hour Terminate

Task 1 ✓ Introduction

Task 2 ✓ A Bit of Background on Linux

Task 3 ✓ Interacting With Your First Linux Machine (In-Browser)

Task 4 ✓ Running Your First few Commands

* Docum

* Manag

* Suppo

System

System

Usage

Memory

Swap u

* Ubuntu

compl

https

Expanded

0 update

Enable E

See http

Conclusion

In this task, we successfully set up a TryHackMe account, configured a VPN connection, and verified connectivity. These steps allowed seamless access to TryHackMe's environment for hands-on exercises. All steps were completed as outlined, with screenshots providing visual confirmation.